CCL6546-K26

Flow Like a Pro Lab Overview

Lab Guide for CCL6546 - Flow Like a Pro

Welcome

Welcome to Flow Like a Pro! In this hands-on lab, we'll dig into how Flow Designer really handles data behind the scenes. You'll learn when pill values refresh (and when they don't), how GlideRecord data can quietly go stale, and how to debug issues that slow you down. By the end, you'll have a stronger grasp of how to build reliable, real-time flows that do exactly what you intended — no surprises.

Estimated Duration: ~65 minutes (including time for Q&A / discussion)

Agenda

- Set Up Your Lab Environment (~5 min)
- **Section 1:** Data Pill Behavior (~15 min)
- **Section 2:** GlideRecord Behavior: Subflow vs. Action (~20 min)
- **Section 3:** Runtime Values and Debugging (~20 min)
- **Wrap-Up & Discussion** (~5 min)

Set Up Your Lab Environment

Estimated Time: 5 minutes

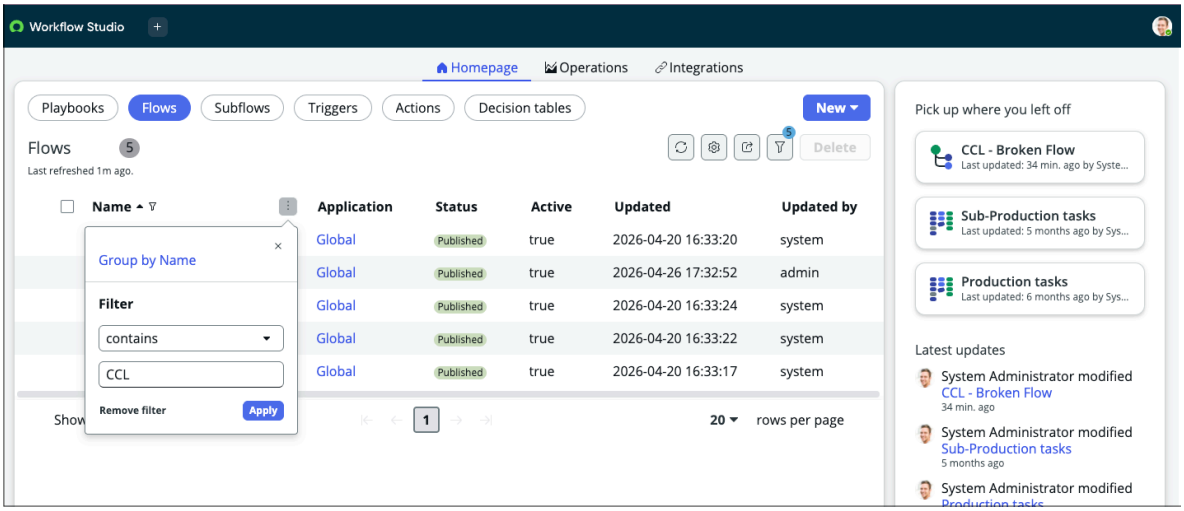
1. **Log in to your assigned lab instance.**

2. **In the application navigator, type Flow Designer and press Enter.**

3. **Confirm that Flow Designer opens and you can see the home screen.**

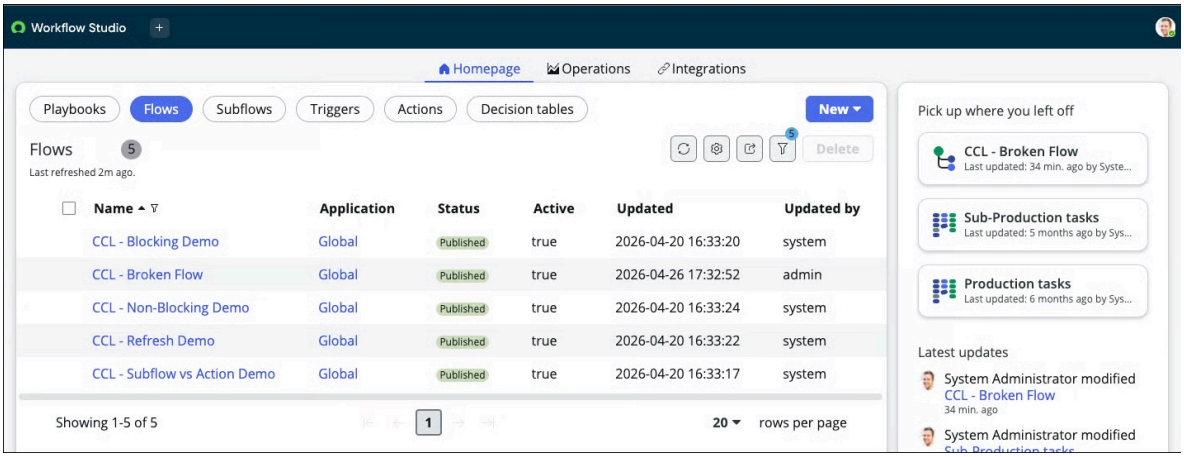
4. **Confirm that the prebuilt flows and subflow are listed:**

   Use the *Name* filter to search for flows with "CCL" in the flow name.



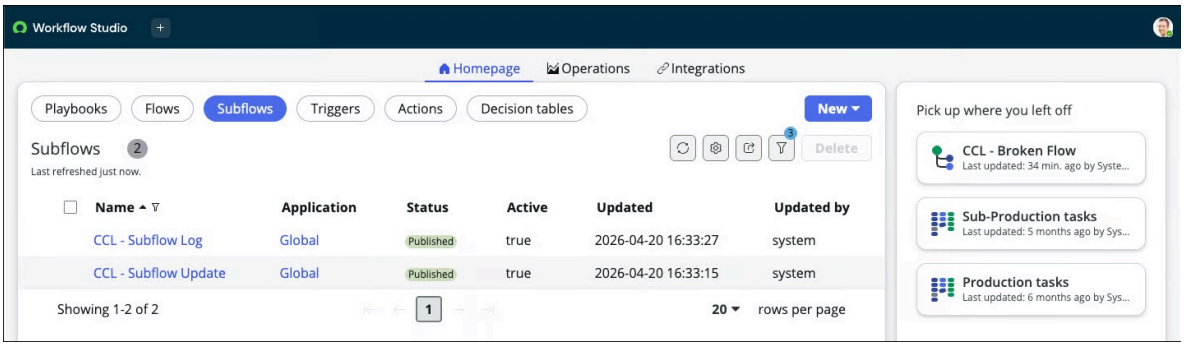
Using the '*Name*' filter in the Flows listing

- Flows
  - **CCL - Blocking Demo**
  - **CCL - Breaking Flow**
  - **CCL - Non-Blocking Demo**
  - **CCL - Refresh Demo**
  - **CCL - Subflow vs Action Demo**



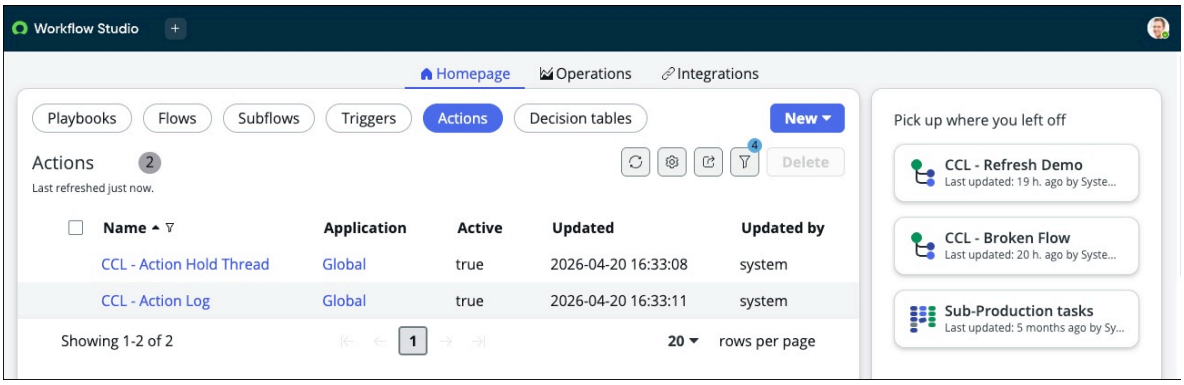
The flows in this lab

- Subflows
  - **CCL - Subflow Log**
  - **CCL - Subflow Update**



The subflows in this lab

- Actions
  - **CCL - Action Hold Thread**
  - **CCL - Action Log**



The actions in this lab

If you don't see the flows, confirm you are logged in with the correct lab credentials and have the `flow_designer` role.

Section 1: Data Pill Behavior

In this section, you'll build and run flows that demonstrate how data pills behave differently depending on whether instructions are blocking or non-blocking, and see how subflows change data behind the scenes.

Estimated Time: 15 minutes

Exercise 1.1: Blocking vs Nonblocking Instructions - Data Pill Behavior

Objective: Understand how data pill values behave when flow actions use blocking vs. non-blocking instructions, and observe that pills reflect the value at the time they were resolved.

Estimated Time: 20 minutes

Context

In Flow Designer, a **blocking** instruction pauses execution until it completes before moving to the next action.

A **non-blocking** instruction kicks off work but allows the flow to continue immediately. This distinction directly affects when your data pills get their values — and whether downstream actions see updated data.

Actions — Blocking Demo

1. **Open the flow "CCL - Blocking Demo" in Flow Designer.**

2. **Review the flow structure. Notice it has the following actions:**

Action	Type	Description
Trigger	Record Created	Fires when an Incident is created where *Short Description = CCL - Blocking Demo*
1	Log Message	Logs the short description value of the trigger Incident record
2	Subflow (Blocking): CCL - Subflow Update	Calls **CCL - Subflow Update** — updates the short description on the Incident to append a new line "Updated by subflow". Runs with *wait_for_completion = true*.
3	Action: CCL - Log Priority	A custom action that receives the trigger Incident record and logs its short description value.
4	Subflow: CCL - Subflow Log	Calls a second subflow that takes the trigger Incident record as input and logs "CCL - Subflow log" + the short description value.

3 **Before running the flow, predict:**

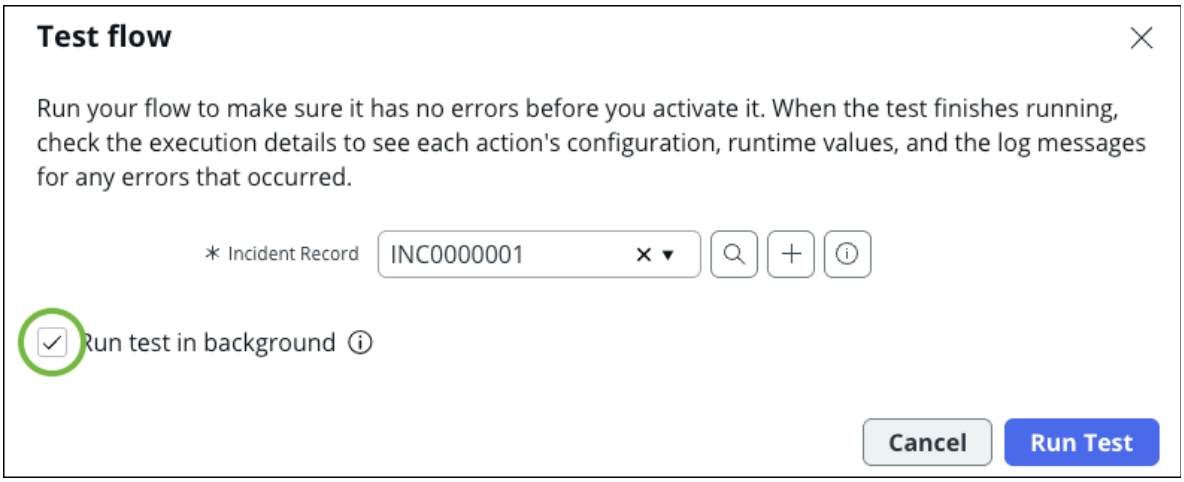
The subflow in Action 2 adds a new line to the short description. Action 2 runs as blocking, so it completes before Action 3. Will the **CCL - Action Log** action in Action 3 show the **original** short description or the **updated** short description with a new line?

4 **Think about it:**

Action 3 is a custom action (not a subflow). How does an action receive the record — as a fresh database fetch or an in-memory object?

5 **Click Test to run the flow.**

Use an existing Incident record as input, select *Run test in the background*, and click **Run Test**. In this example 'INC0000001' is used.

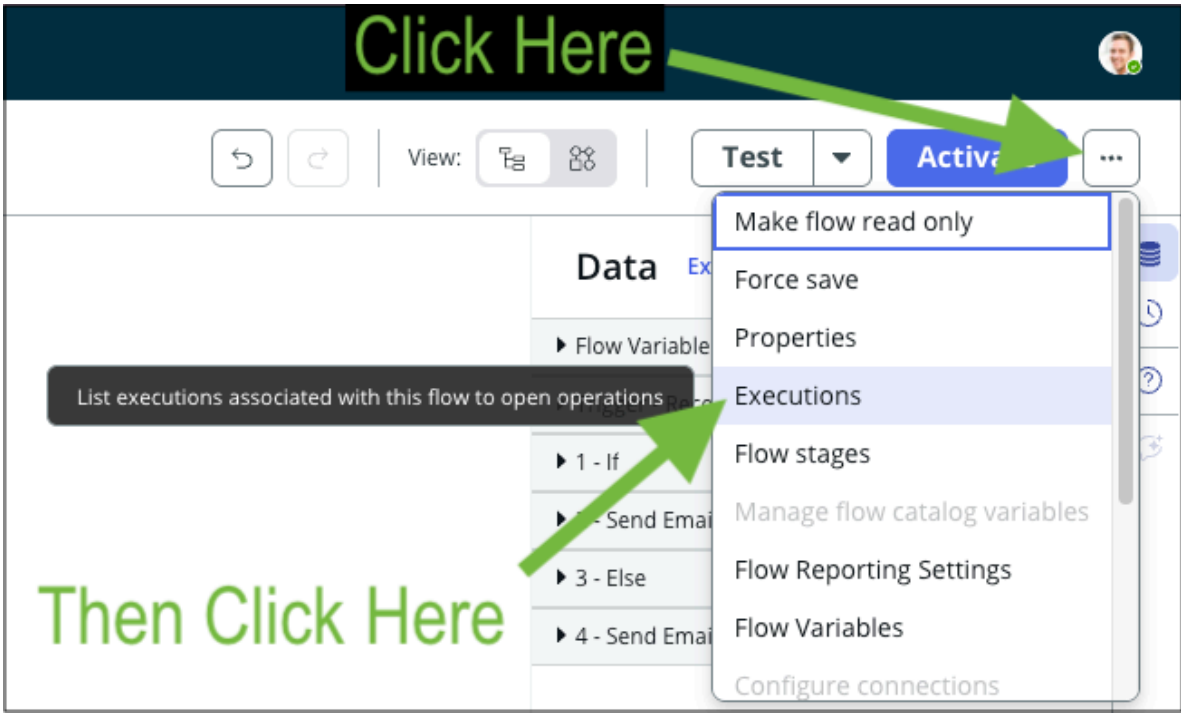


Test

6 **Once the flow completes, open the Execution Details.**

Close the Test dialog box. Click on the **More Actions menu** button to the right of the **Activate** button. The **More Actions menu** button is identified with ellipsis.

Select the **Executions** option to open the list of executions for this flow and then select the most recent entry.



More Actions menu > Executions

7

A note about testing flows with the Test button:

The Test button in Flow Designer runs a flow either in the foreground or background, controlled by the *Run test in background* checkbox. This distinction matters more than it might seem.

Running a flow **in the foreground** executes it within your current user session, which can introduce subtle differences in behavior — the flow may not run as the correct user or reflect the same conditions as a real trigger. Running **in the background** ensures the flow executes as the intended user and behaves exactly as it would during a normal trigger event.

The flows in this lab — such as **CCL - Subflow vs Action Demo** — have their trigger defined as *Run flow in background***.** To match that trigger definition when using the Test button, make sure the *Run test in background* checkbox is checked.

One final note: the Test button **bypasses trigger evaluation entirely.** To test whether trigger conditions are working correctly, the flow must be activated and a record manually created or updated to satisfy the trigger conditions.

8 Compare the log outputs from Action 1 and Action 3.

Action 1

Action 3, with 'Steps' expanded

When *Wait for Completion* is checked, you are using a blocking instruction - the subflow runs synchronously on the same thread as the main flow.

After the subflow finishes and commits its changes to the database, the Flow Engine automatically re-fetches every record input and writes the refreshed GlideRecord back into the flow's runtime values. This refresh happens once the subflow completes, before the next action runs.

This is why action 3 (Action Log) and action 4 (Subflow Log) both display the updated short description value - the main flow's in-memory record was replaced with a fresh DB copy the moment the subflow returned.

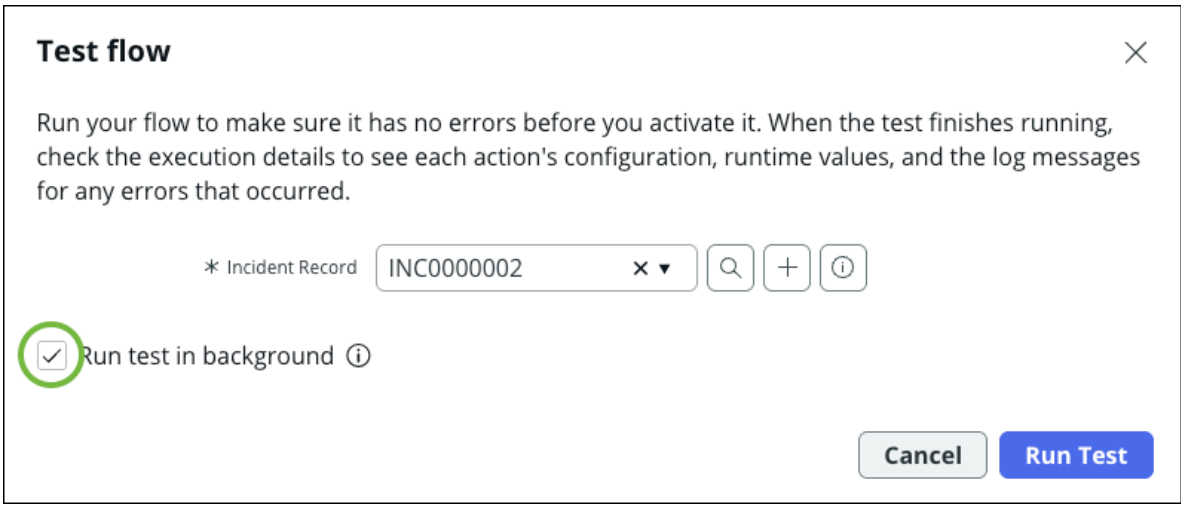
Actions — Non-Blocking Demo

1 **Now open the flow CCL - Non-Blocking Demo.**

This flow has the same structure, except **Action 2 calls the subflow as non-blocking** *wait_for_completion = false*.

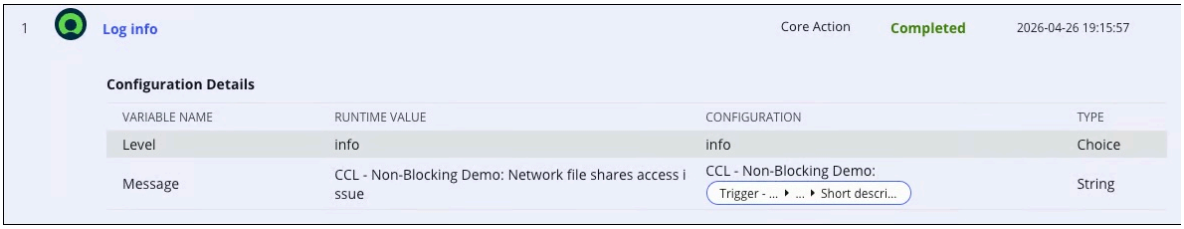
2 **Run the test again, this time with INC0000002**

Repeat the testing procedure from the previous example.

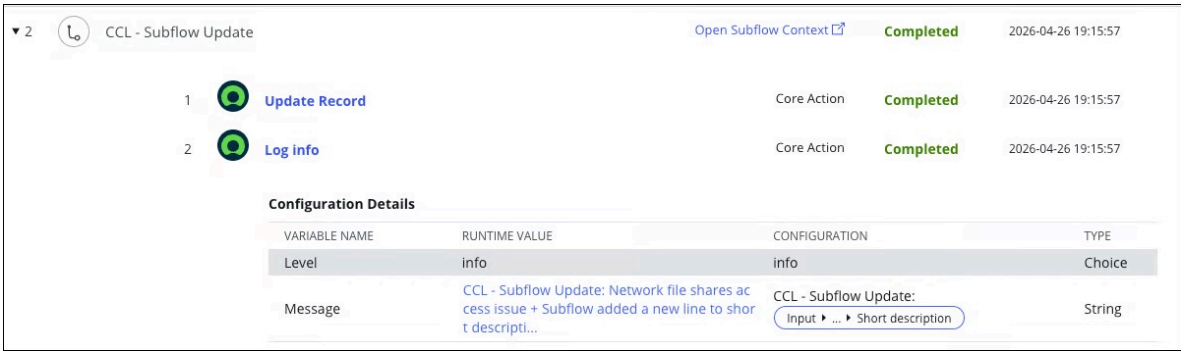


Test Dialog

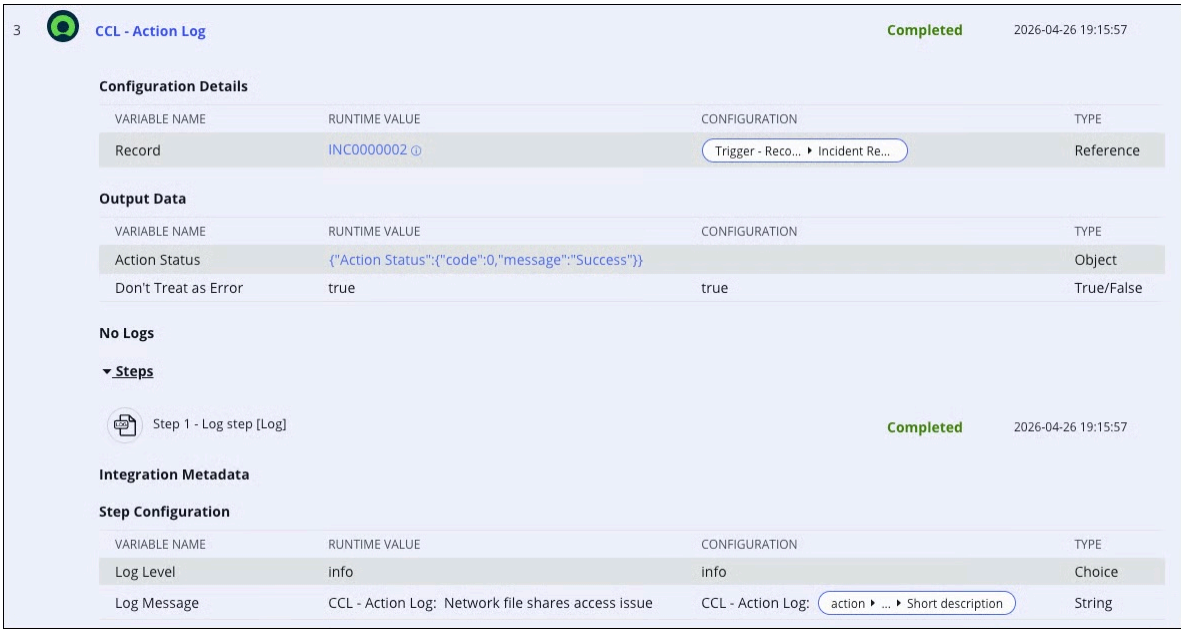
3 Compare the execution details.



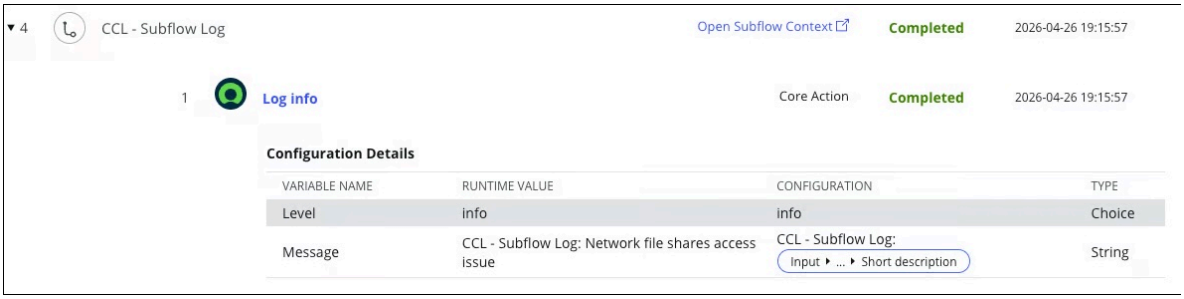
Action 1



Action 2, Subflow



Action 3, Custom Action



Action 4, Second subflow

When *Wait for Completion* is unchecked, the subflow is scheduled as a separate job on a different thread and transaction. The main flow does not wait for it, and the engine does not refresh the main flow's runtime values afterward.

As a result, any downstream action in the main flow keeps using the original in-memory GlideRecord and logs the original short description — even if the async subflow has already committed its update to the database. A downstream subflow still performs its own DB fetch, so it may or may not see the update depending on whether the async subflow has finished writing.

Takeaway

The *Wait for Completion* field determines whether an instruction is blocking or nonblocking, and this affects data availability to every downstream step.

Blocking: The subflow runs synchronously on the same thread. When it finishes, the engine re-queries every record input from the database and replaces the copy in the main flow's runtime values. Downstream actions and subflows will **see the updated record.**

Non-blocking: The subflow is queued on a separate thread/transaction; no refresh happens. Downstream actions see the **original in-memory record.** Downstream subflows will do their own DB fetch, so they see whatever is in the DB when they run.

Section 2: GlideRecord Behavior - Subflow vs. Action

In this hands-on session you'll observe a key difference in how GlideRecord data is passed to a subflow versus an action.

Estimated Time: 20 minutes

Exercise 2.1: Compare GlideRecord Values - Subflow vs Action

Objective: Understand that when you pass a GlideRecord to a subflow, it receives a reference (sys_id + table) and re-fetches from the database, while an action within the same flow uses the already-loaded in-memory record object — and how this affects data freshness.

Context

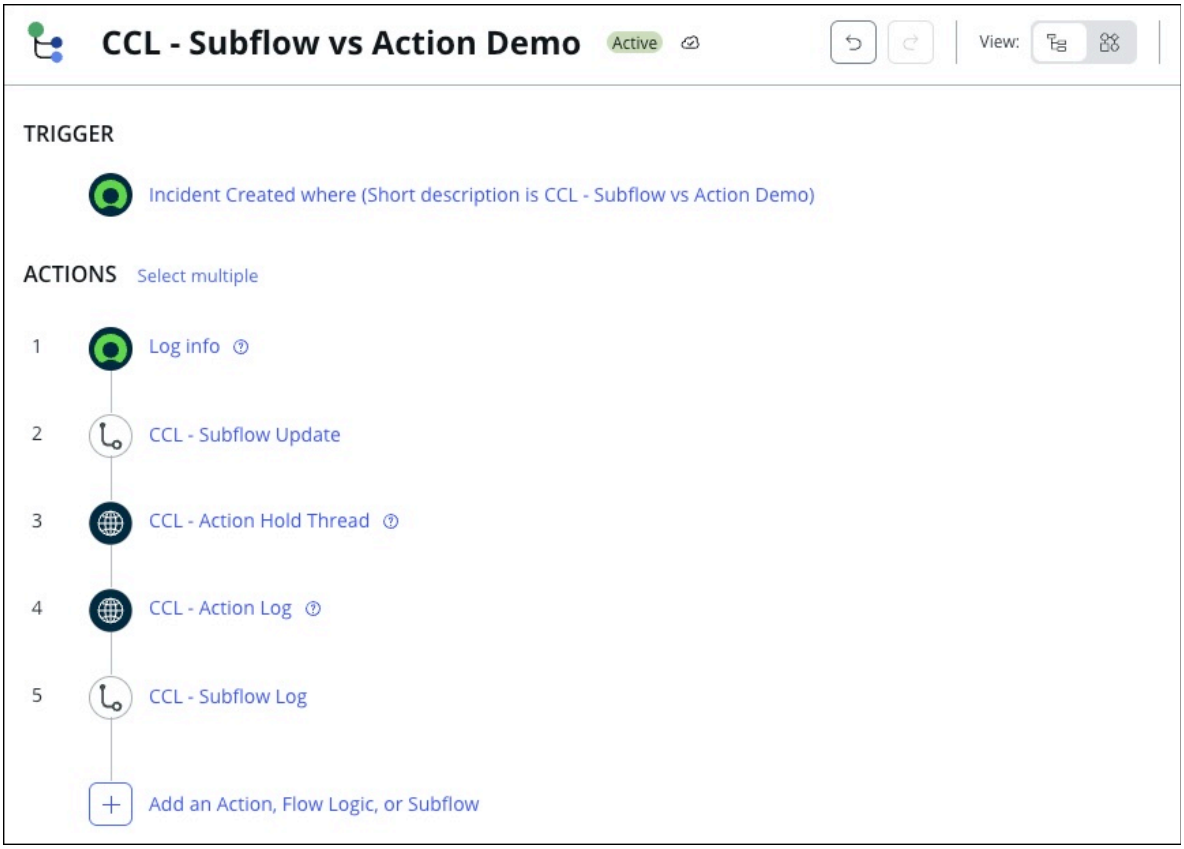
When you pass a record pill to a **subflow**, the Flow Engine makes a call to retrieve a fresh GlideRecord from the database. The subflow always starts with the **current DB state.**

When the same pill is used in an **action** within the same flow, the engine does a simple map lookup in the flow's runtime values and hands the action the **in-memory GlideRecord object.** The action does **no DB query**; it sees whatever the most recent refresh left in memory... meaning it could reflect stale data if the record was modified after the original query.

Walkthrough

1

Open the flow "CCL - Subflow vs Action Demo".



CCL - Subflow vs Action Demo

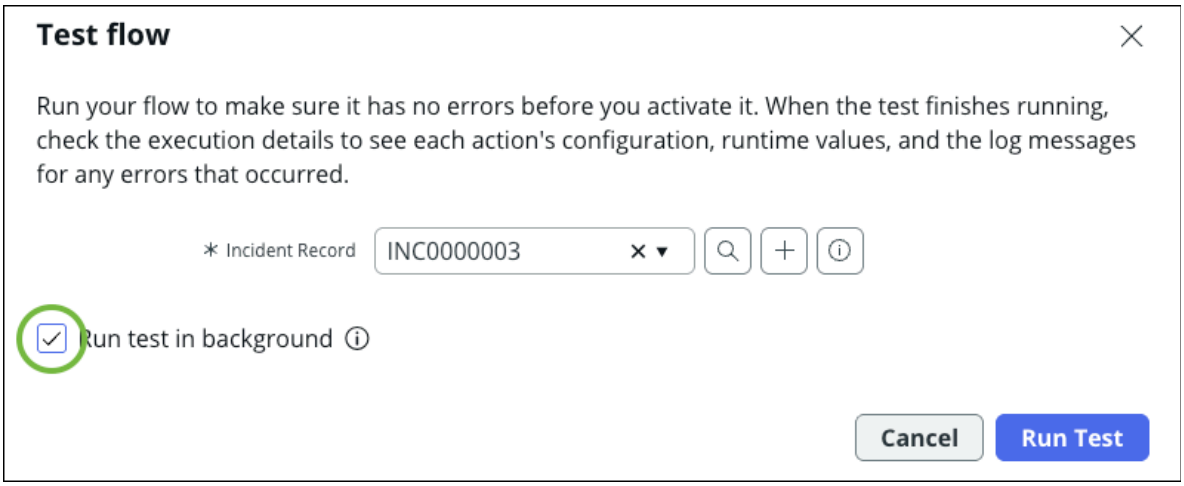
2 **Review the structure:**

Action	Type	Description
Trigger	Record Created	Fires when an Incident is created where *Short Description = CCL - Subflow vs Action Demo*
1	Log Message	Logs the initial short description of the trigger Incident record
2	Subflow: CCL - Subflow Update	Calls **CCL - Subflow Update** — updates short description to add a new line "Updated short description." Runs with *wait_for_completion = false*.
3	Action: CCL - Action Hold Thread	A custom action that **pauses execution for 10 seconds**, giving the non-blocking subflow enough time to complete
4	Action: CCL - Action Log	A custom action that takes the trigger Incident record and logs its short description
5	Subflow: CCL - Subflow Log	Calls a subflow that takes the trigger Incident record and logs its short description

3

Click Test to run the flow

This example will use INC0000003.

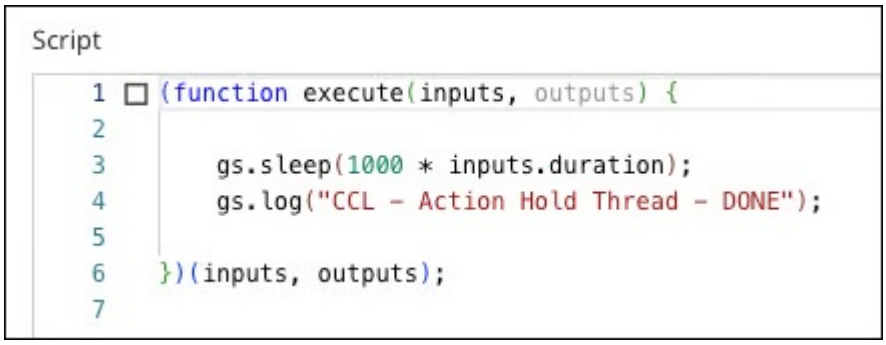


Run the test in the background

4

Why the 10-second wait?

Action 2 fires the subflow as non-blocking, so the flow continues immediately.
Action 3 pauses for 10 seconds to ensure the non-blocking subflow has time to finish updating the database before we check the results.

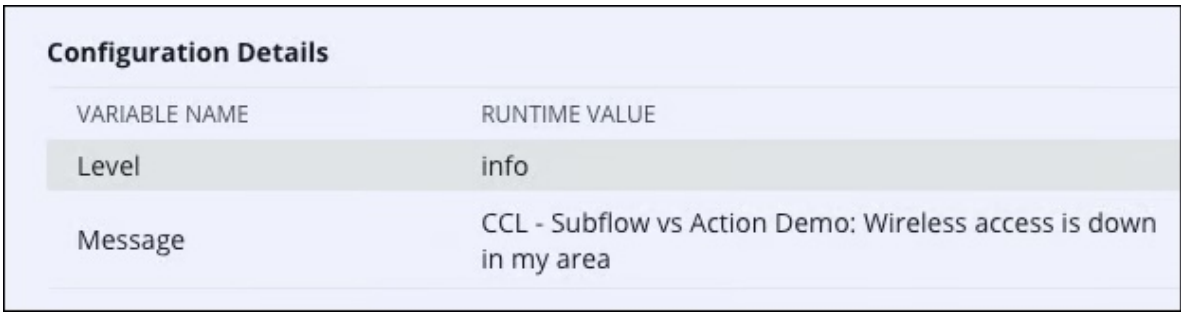


The script in the custom action

5 **Observe the results:**

> Wait 10 seconds after clicking 'Run Test' as the flow is still executing in the background even after the dialog closes. Then open the executions for this flow and open the most recent one.

- **Action 1 (Log):** Logs the **original** *short description* — before any changes.

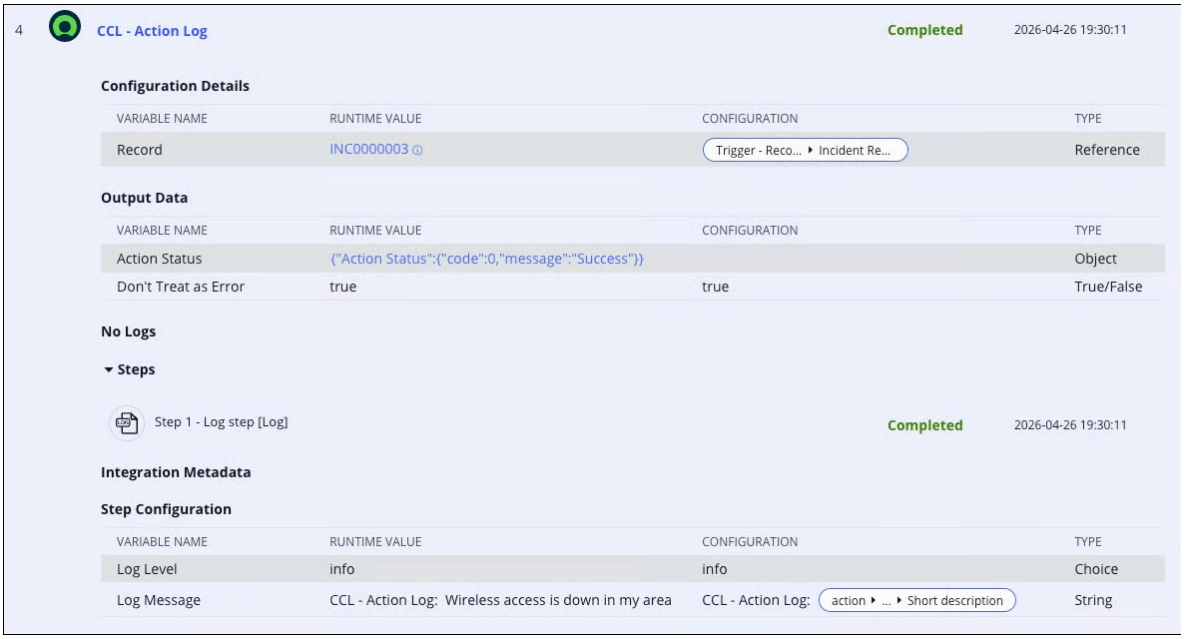


Configuration Details

VARIABLE NAME	RUNTIME VALUE
Level	info
Message	CCL - Subflow vs Action Demo: Wireless access is down in my area

Action 1 Detail

- **Action 3 (Action: CCL - Action Hold Thread):** This action uses `gs.sleep` to 'pause' the execution. This is an anti-pattern used for demonstration purposes only. `gs.sleep` does not put the flow to sleep, it is a busy wait, continuing to hold the active thread and preventing any other work to happen in it.
- **Action 4 (Action: CCL - Action Log):** Even after waiting 10 seconds for the subflow to complete, this action still logs the **original** short description.
  Why? Because it's an action that uses the **in-memory GlideRecord** from the trigger — it never re-fetches from the database.



Action 4 Detail

- **Action 5 (CCL - Subflow Log):** Because subflows receive a reference (table + sys_id) and **re-fetch the record from the database**, this subflow sees the **updated** short description.

Action 5 Detail

6

Why does this matter?

This is one of the most common sources of "my flow isn't working" confusion. The 10-second wait proves the subflow had plenty of time to complete — yet the action **still** shows stale data.
The issue isn't timing; it's **how the record is passed (action vs subflow).**

Key Differences Summary

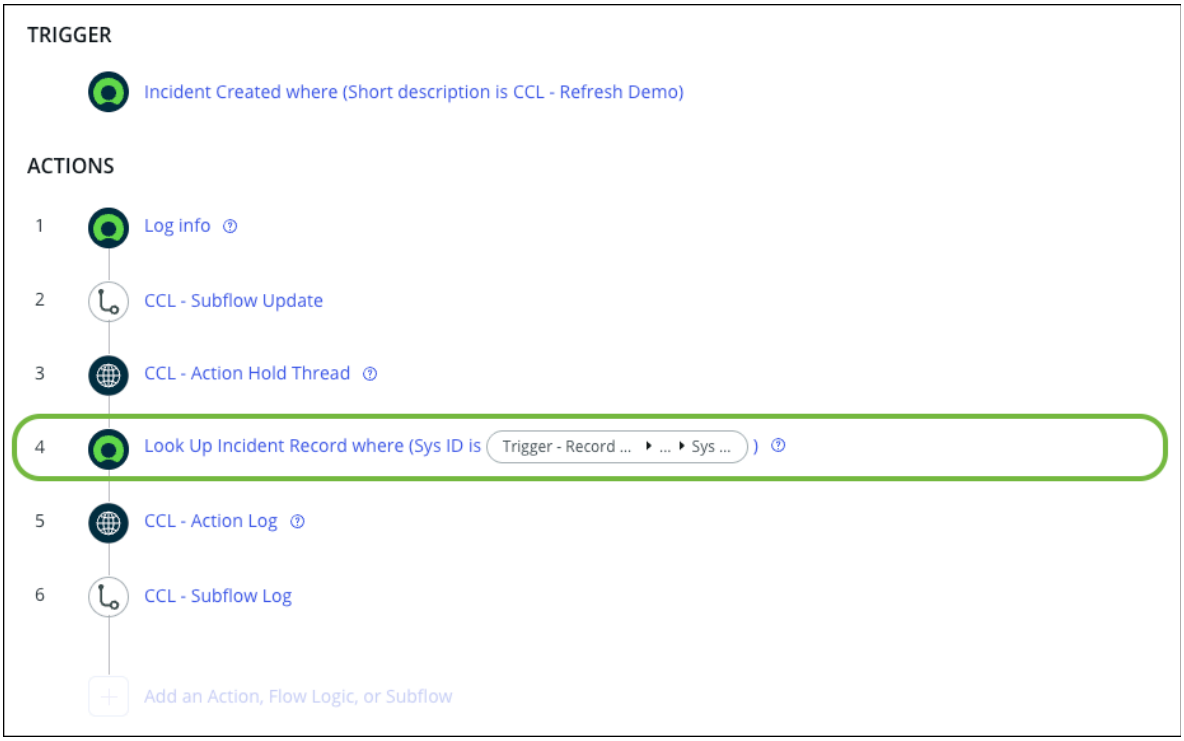
	Subflow	Action
Record passed as	Fresh GlideRecord	In-memory GlideRecord object
Data freshness	Re-fetched from DB — always current	Uses cached in-memory value — may be stale
When to watch out	Generally safe for fresh data	Be cautious if record was modified elsewhere

Exercise 2.2: Use Look Up Record to Refresh Stale Data

1

Open the CCL - Refresh Demo flow.

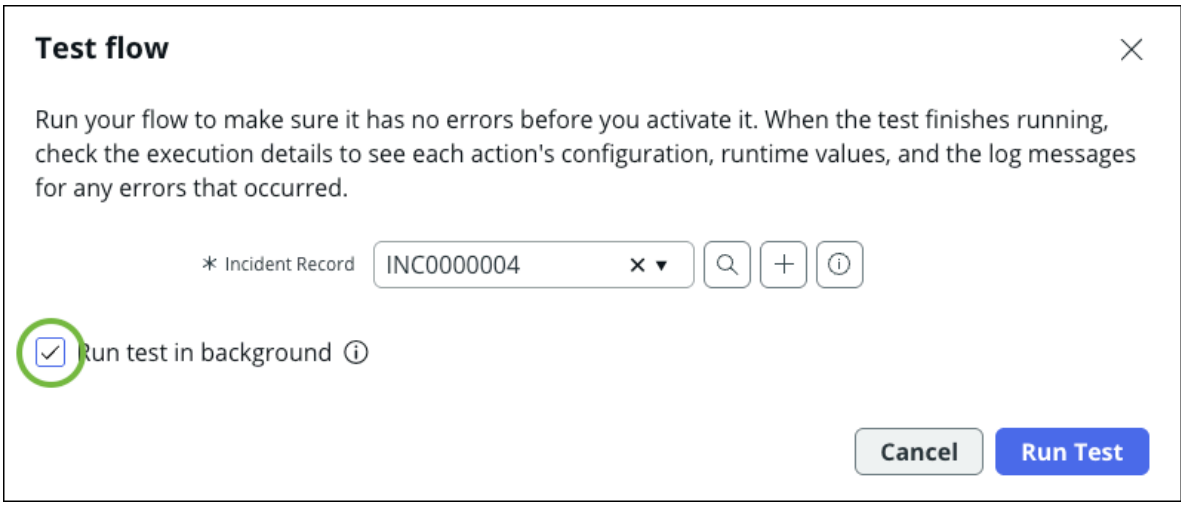
Notice the additional **Look Up Record** action after the subflow, querying the same Incident by its Number.



An explicit lookup record

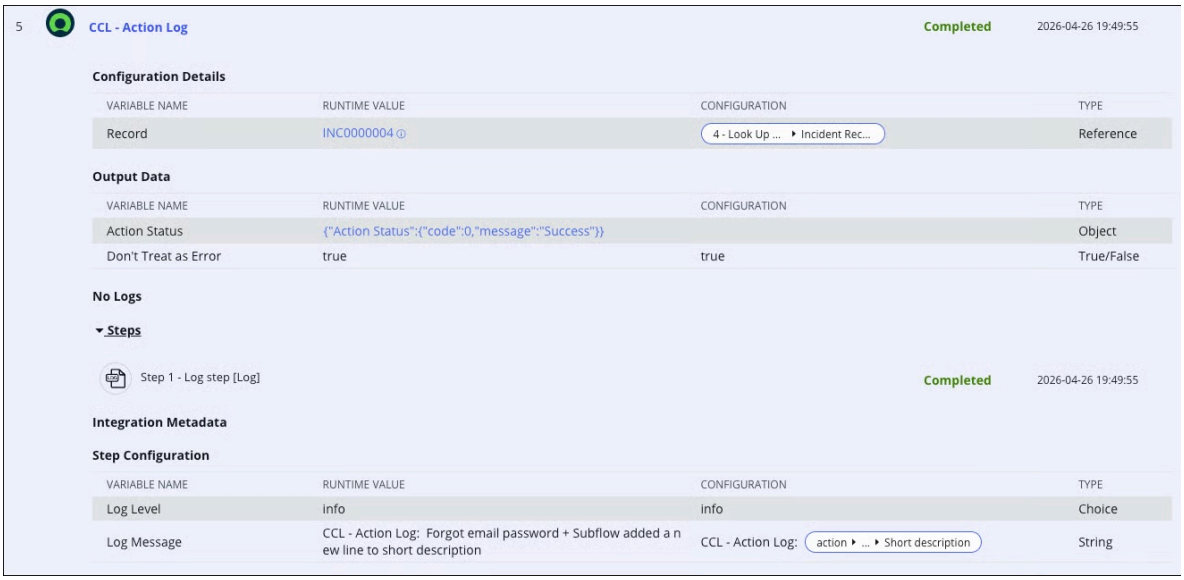
2

Run the test with INC0000004.



Run the test in the background

Now, the last log reflects the new updated short description value.



Example execution

The third log now shows the updated short description — because the Look Up Record performed a fresh database fetch.

Takeaway

When you need guaranteed fresh data in a downstream action, prefer passing the *sys_id* and using **Look Up Record** within that action to re-fetch, rather than relying on how the record pill is implicitly passed.

Section 3: Runtime Values and Debugging

Objective: Learn to launch Debug mode in Flow Designer, set breakpoints, and use the debugger to inspect runtime values step by step.

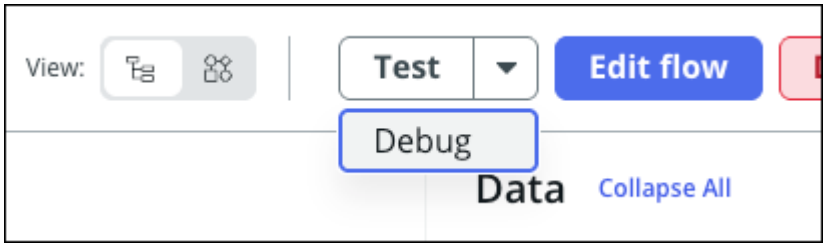
This section covers where runtime data lives and how to access it for debugging.

Estimated Time: 20 minutes

3.1: Launching Debug Mode

1. **Open the "CCL - Blocking Demo" flow from the earlier exercises.**

2. **Click the dropdown arrow next to the Test button and select Debug.**



The 'Debug' button is under the ▼

3. **Enter your trigger inputs.**

The input interface is the same as Test mode. For this example use INC0008112.

Test with INC0008112

4

Click Debug. The flow opens in a new Workflow Studio tab in Debug Mode.

Input records use their current state, not their initial state — same behavior as Test mode.

> No special role is required beyond Flow Designer access. If you can open Flow Designer, you can use the debugger. The Script Debugger role is only needed when stepping into the platform script debugger.

3.2: Navigating with Debug Controls

Use the five navigation controls in the top right of the debug tab to step through execution:

The five debugging controls

1. **Resume:** Runs until the next breakpoint (or to completion if none remain)
2. **Step Over:** Advances one step regardless of breakpoints; treats subflows as a single step
3. **Step Into:** Opens a subflow or script in its own debug tab; parent flow pauses
4. **Step Out:** Exits the current subflow and returns to the parent tab
5. **Skip All Breakpoints:** Runs the flow to completion, ignoring remaining breakpoints

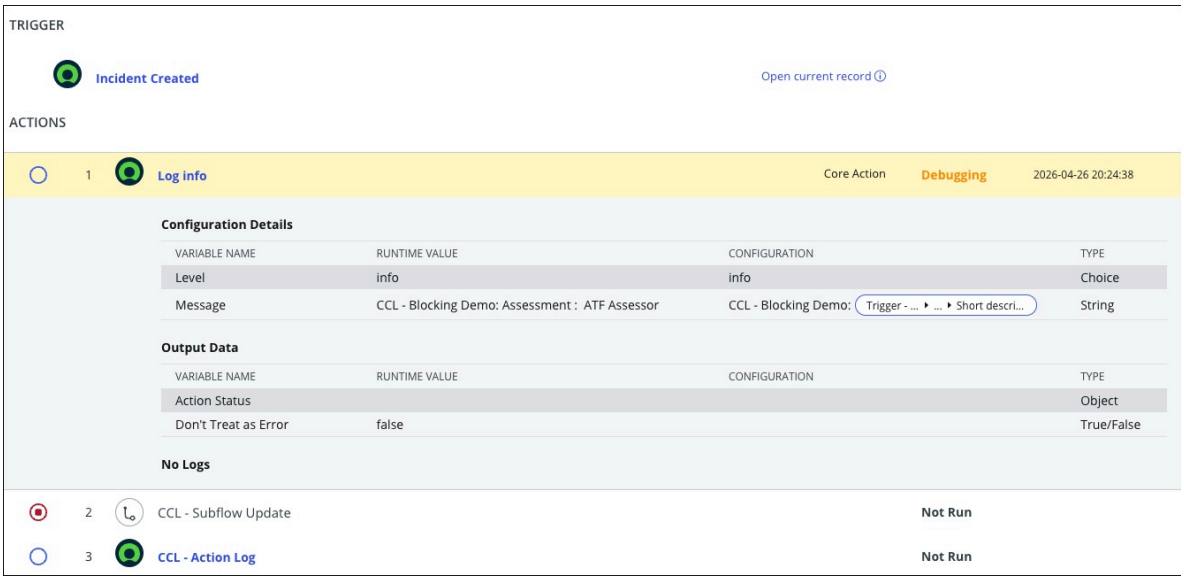
For **custom action scripts**: When paused at a script step, click **Step Into** (or the **Script Debug** inline link on the step) to open the platform script debugger. Step through the script line by line and inspect variables. Script outputs are surfaced back in the Flow Debugger after completion. **Requires the Script Debugger role.**

For **loops**: Set a breakpoint **inside** the loop body. Each **Resume** advances one iteration — inspect data after each pass. Once the early iterations look clean, remove the loop breakpoint and **Resume** to run through the rest.

3.3: Setting Breakpoints & Inspecting Runtime Values

1 **In the debug tab, click the circle icon to the left of any action to set a breakpoint.**

The circle fills in to indicate it is active.



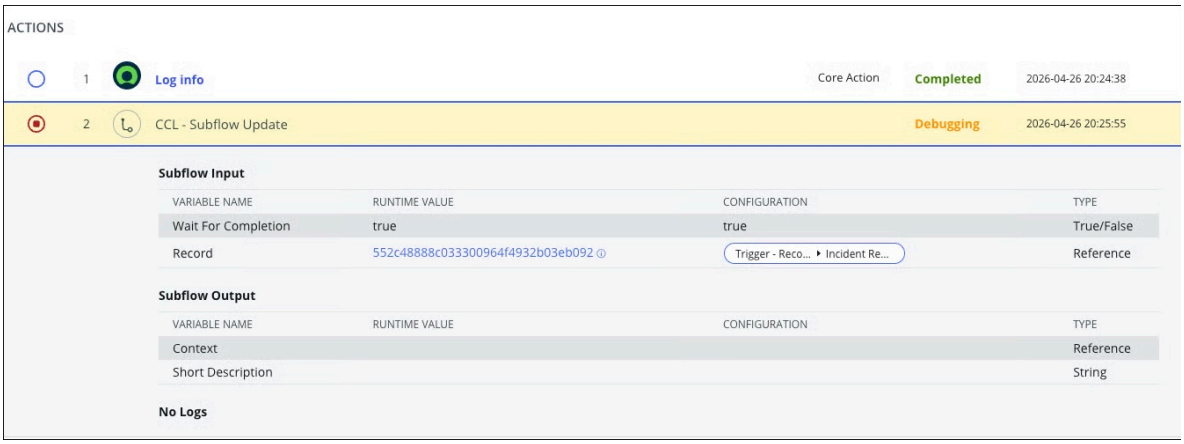
Initial debugging view with a breakpoint on action 2

Good candidates for strategic breakpoints are subflow calls, conditional branches, and actions that write to records.

2

Click Debug / Resume to run the flow.

Execution pauses at the first breakpoint and an orange "Debugging" badge appears on the active action.



Debugging paused on action 2

3

Review the Runtime & Configuration Values panel for the paused action.

- *Variable Name*: The variable from this action
- *Runtime Value*: What the action actually produced at runtime
- *Configuration*: What you configured (pill references, static values)
- *Type*: Data type (Reference, String, True/False, etc.)
- Compare Runtime Values to Configuration to identify mismatches.

> You can add or remove breakpoints at any time during an active session — no need to restart.

3.4: Hands-On — Debug the CCL - Broken Flow

Scenario: The **CCL - Broken Flow** should send a tailored email to the assigned engineer whenever an incident is created or updated. If the incident is urgent, the engineer's manager should be CC'd and the email should request prioritization. If not urgent, a standard notification goes to the engineer only.

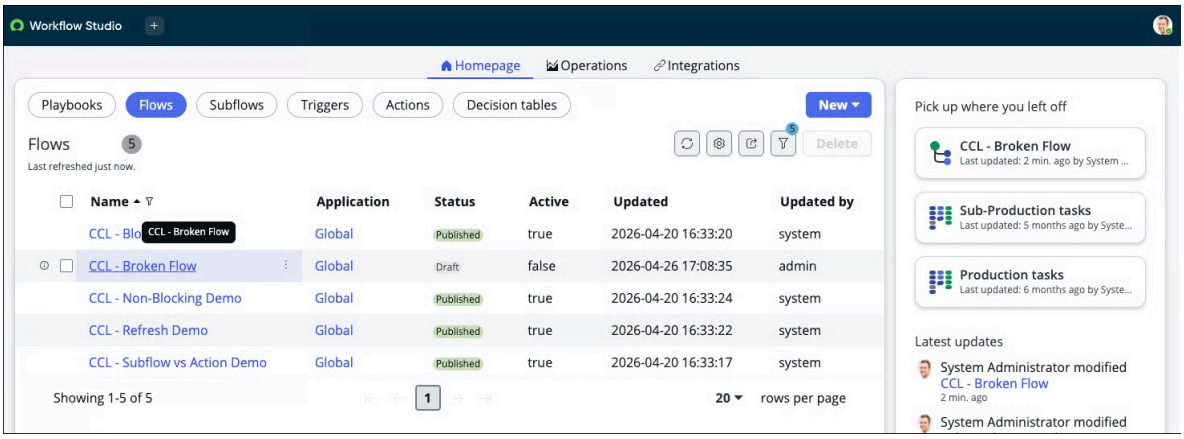
Three problems have been reported: emails sometimes aren't sent at all, the wrong email type goes out on genuinely urgent incidents, and when emails do attempt to send, they fail to deliver.

You have two real incidents to work with:

	INC0000054	INC0000049
Urgency	1 — High	1 — High
Priority	1 — Critical	2 — High
Assigned to	*(empty)*	Don Goodliffe
Surfaces	Bug 1	Bug 2 + Bug 3

3.4.1 — Find Bug 1 using INC0000054

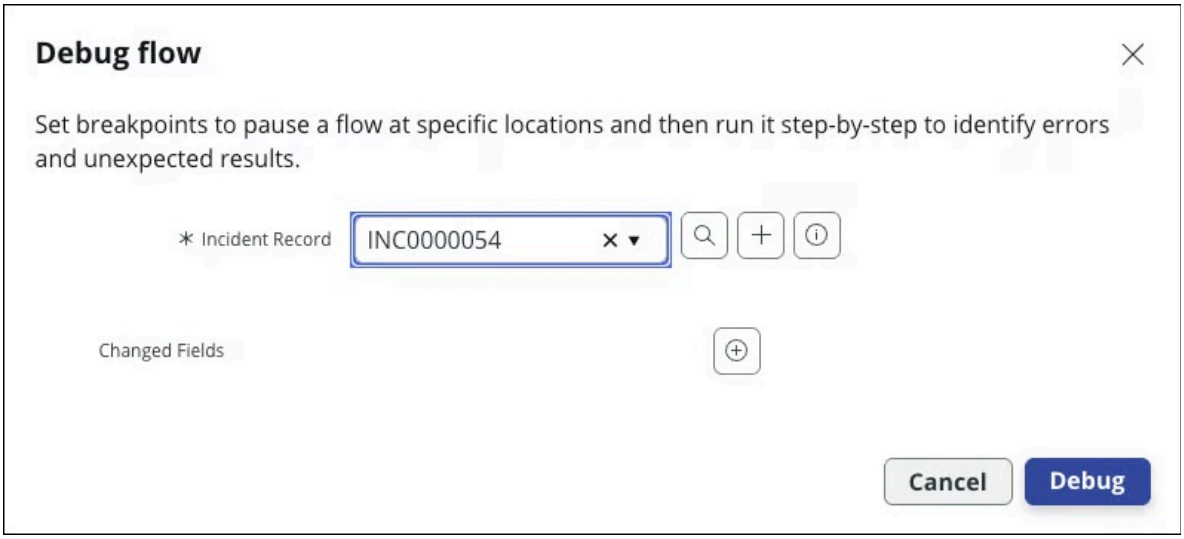
1 Open CCL - Broken Flow in Workflow Studio



Select the CCL - Broken Flow

2 Click Test ▼ > Debug

1. Select **INC0000054** as your trigger record
2. Click **Debug**.

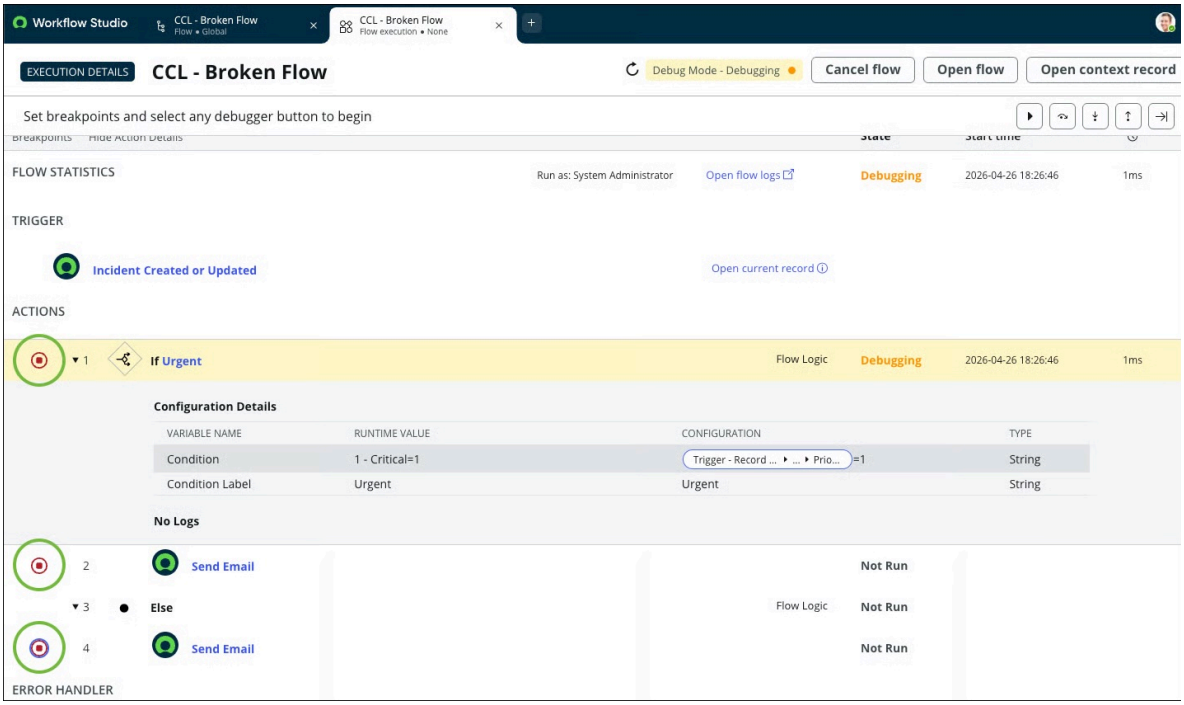


Debug with INC0000054

3

Set breakpoints on:

- Action 1 — the **If Urgent** condition
- Action 2 — the **then** Send Email (Urgent Incident Assigned)
- Action 4 — the **else** Send Email (New Incident Added)



Add breakpoints on all actions in the flow

4

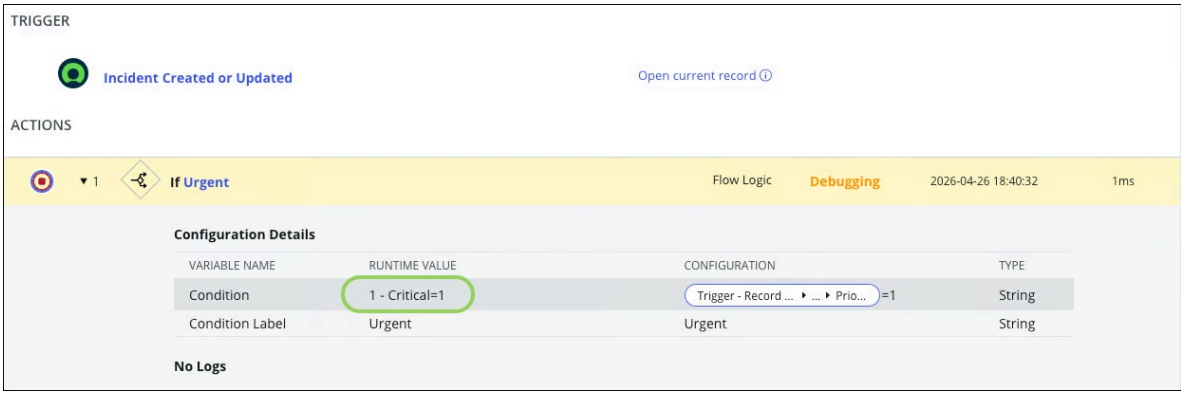
Notice!

The execution is already paused at Action 1.

5

Expand the Runtime & Configuration Values panel

- What field does the **Configuration** column show for Action 1?
- What is the **Runtime Value**?



This is a display issue only, the condition is successfully evaluated

INC0000054 has *Priority 1 — Critical*, so the condition evaluates true and the flow takes the then branch. Note the field being checked — we'll come back to this in Part 2.

The display here is showing a mix between Display Value and Value for the configuration panel.

6

Click Resume

Execution pauses at Action 2 — the urgent Send Email.

7

Expand the Runtime & Configuration Values panel on Action 2

- Find the **To** field. What is its **Runtime Value**?
- **Expected finding**: *null*. INC0000054 has no Assigned To populated.

8

This is Bug 1

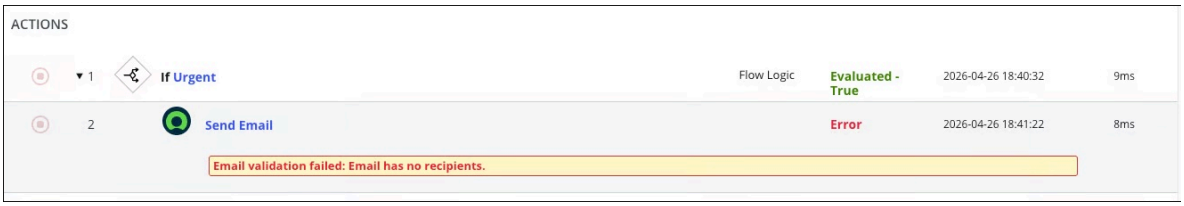
The flow has no guard checking that Assigned To is populated before attempting to send. With a null recipient, the Send Email action will error or send to nobody.

Discussion: Where should the guard be added, and what should it check?

Answer: An If condition before the branching logic — `Trigger > Record > Assigned to is not empty`. The entire email logic sits inside the then branch. The else branch takes no action.

9

Click Skip All Breakpoints to end the session cleanly



The final state of the flow execution

3.4.2 — Find Bugs 2 and 3 using INC0000049

1. **Launch a new debug session**

   Select **INC0000049** as your trigger record.

   - INC0000049 has *Urgency 1 — High* but *Priority 2 — High*. It is genuinely urgent. It has an assignee. The flow should take the urgent path and CC Don Goodliffe's manager.

2. **Set breakpoints on actions 1, 2, and 4**

3. **Execution is already paused at action 1**

   Expand the Runtime Values panel on the If condition

   - **Configuration** column: what field is the condition checking?
   - **Expected finding**: *Trigger - Record > Priority = 1*
   - **Runtime Value** for that field?
   - **Expected finding**: *Priority = 2 — High = 1*.

Paused on Action 1

   - Click **Resume**. Note the flow takes the **Else** branch despite Urgency 1 — High.

4

This is Bug 2

- The condition checks *Priority* instead of *Urgency*. Click **Resume** — execution pauses at Action 4, the standard **Send Email**.

5

Expand the Runtime Values panel on Action 4

Locate the **To** field.

- **Runtime Value:** what does it show?
- **Type:** what data type is shown?
- **Expected finding**: Runtime Value shows a *sys_user* record reference — something like a *sys_id* or a user object. Type shows **Reference**, not **String**.

6

This is Bug 3

The To field data pill is configured to resolve to the Assigned To user *record* — a Reference to the sys_user table — rather than drilling into that user's **Email** field. The Send Email action needs a string email address, not a Reference object. The same misconfiguration exists on the CC field (Manager) in the urgent branch.

> **The distinction:** The data pill *Trigger > Record > Assigned to* resolves to the user record. The correct pill is *Trigger > Record > Assigned to > Email* — one level deeper, resolving to the email address string on that user record.

Assigned To is evaluated as a sys_id

7 **Click Skip All Breakpoints to end the session**

Debrief — all three bugs:

Bug	Test incident	Where found	What Runtime Values showed
Bug 1 — null recipient guard	INC0000054	Action 2 To field	Runtime Value = null
Bug 2 — wrong condition field	INC0000049	Action 1 condition	Configuration = Priority; wrong branch taken
Bug 3 — wrong data pill depth	INC0000049	Action 4 To field	Runtime Value = Reference object, Type = Reference not String

> Bug 1 and Bug 2 would have been hard to diagnose from logs alone. Bug 3 might never appear in logs at all — the flow completes, an email is attempted, and it silently fails to deliver because the recipient field contained a record reference instead of an address. The Type column in the Runtime Values panel surfaces this in seconds.

3.5 Challenge — Fix and Verify

1

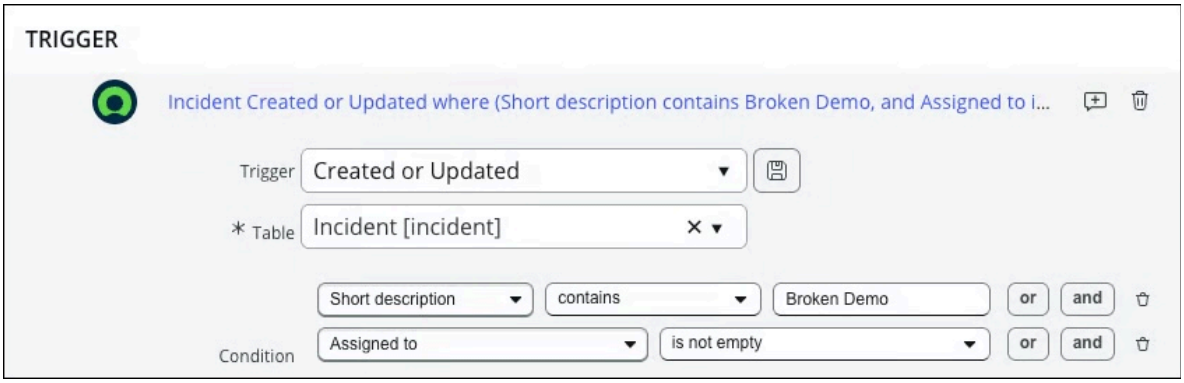
Open the Flow CCL - Broken Flow

- Select the tab in workflow studio.
- click **Edit Flow**.

2

Fix Bug 1

Update the trigger condition to include: *Trigger > Record > Assigned to is not empty*.



Bug 1 fix

3

Fix Bug 2

Open Action 1 (If Urgent). Change the condition field from *Priority* to *Urgency*, operator *is*, value *1 — High*. Save.

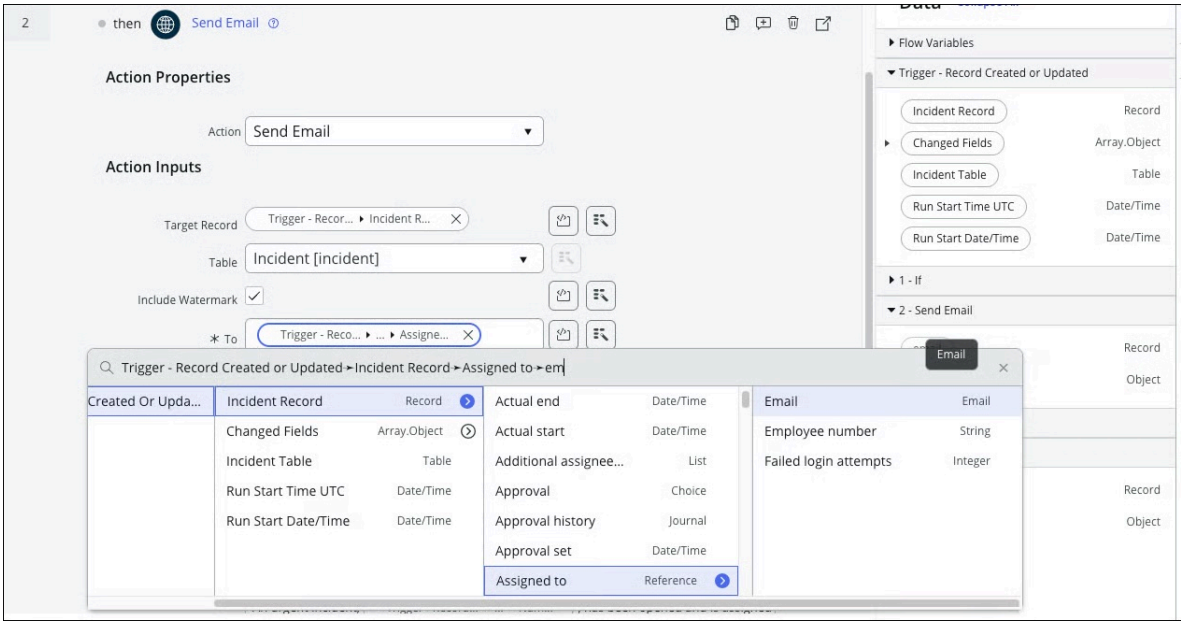
Bug 2 fix

4 Fix Bug 3

Open Action 2 (urgent Send Email). Update the **To** data pill using the right hand data menu:

- Remove the current pill *Trigger > Record > Assigned to*
- Re-add it drilling one level deeper: *Trigger > Record > Assigned to > Email*
- Update the *CC* field the same way: *Trigger > Record > Assigned to > Manager > Email*

Open Action 4 (standard Send Email) and apply the same fix to the **To** field, this time do it by clicking the pill and continuing to drill down: *Trigger > Record > Assigned to > Email*



5 Verify all three fixes using INC0000049

Launch debug mode, select INC0000049.

- Set breakpoints on If Urgent, and the Send Email actions.
- **Resume** → If Urgent: Runtime Value = *1 — High*. Condition *true*, urgent branch taken.
- **Resume** → Action 2 and 4, Send Email: *To* Runtime Value = *an email address string*. Type = String.

Click **Skip All Breakpoints** to end the session

6 Verify using INC0000054

Launch a new debug session, select INC0000054.

- **Resume** → Assigned To guard: Runtime Value = null. Condition false, flow exits cleanly without attempting to send.

Click **Skip All Breakpoints** to end the session

Takeaway

The Flow Debugger replaces the old loop of **run → dig through logs → guess where it went wrong.**
Set breakpoints where you suspect problems, use the **Runtime Values** panel as your source of truth, and **Step Into** subflows and scripts to isolate nested logic.

Wrapup & Discussion

Estimated Time: 5 minutes + open discussion

Key Concepts Recap

- **Data pills are snapshots** — they capture the value at the time their originating action runs and do not auto-refresh.
- **Blocking vs. non-blocking** changes when subflows execute relative to each other, but in-memory record references remain stale regardless of timing.
- **Subflows re-fetch records** from the database (via *table* + *sys_id* reference), while **actions** in the same flow use the cached in-memory GlideRecord — leading to different values for the same record.
- **Look Up Record is your refresh button** — use it whenever you need guaranteed current data after a modification.
- **The CCL - Subflow vs Action Demo isolates the non-blocking use case** — even with a 10-second wait ensuring the subflow completes, the action still sees stale data. The difference is in **how** the record is passed, not **when**.

Discussion Questions

- Have you encountered stale data issues in your production flows? What was the symptom?
- How might you redesign an existing flow to avoid stale pill problems?
- When would you choose to use a subflow vs. an action, knowing how record data is passed differently?
- In the Subflow vs Action Demo, why does adding a wait *not* fix the stale data problem for the action?

Thank you for attending! We hope this lab gives you a clearer mental model of how data flows through Flow Designer. Happy building!

Reference Material

Workflow Center of Excellence:

https://sn.works/coe ↗

Flow Designer Data pill documentation:

https://www.servicenow.com/docs/r/build-workflows/workflow-studio/data-population.html ↗